

SSN College of Engineering – Kalavakkam
Computer Science and Engineering
ICS1513 – Computer Networks Laboratory

Experiment 1 – Analysing network diagnostic tools

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25.06.2026

1a. Apply and analyze the use of various networking diagnostic and monitoring tools in a computer network environment. Demonstrate the execution of the following tools and record your observations regarding network status, configuration, connectivity, routing, and performance:

```
=====
COMMAND 1 / 16: netstat
RUNNING: netstat -tuln
=====

Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.1:33060         0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:631          0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.53:53          0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:27017        0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:3306         0.0.0.0:*               LISTEN
tcp6       0      0 :::1:631                :::*                     LISTEN
udp        0      0 127.0.0.53:53          0.0.0.0:*               *
udp        0      0 0.0.0.0:49555          0.0.0.0:*               *
udp        0      0 0.0.0.0:631            0.0.0.0:*               *
udp        0      0 0.0.0.0:43326          0.0.0.0:*               *
udp        0      0 0.0.0.0:5353           0.0.0.0:*               *
udp6       0      0 :::49866                :::*                     *
udp6       0      0 :::5353                 :::*                     *

=====
[ SCREENSHOT BUFFER: 5 SECONDS REMAINING ]
=====
```

• netstat

netstat (Network Statistics)

Purpose: Displays active network connections, routing tables, and interface statistics.

Observation Focus: The -tuln flags show listening (l) TCP (t) and UDP (u) ports numerically (n). Look for 0.0.0.0:XX or :::XX, which indicates services on your local machine actively waiting for incoming network connections.

```

COMMAND 2 / 16: ifconfig
RUNNING: ifconfig
=====

eno1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.6.13.14 netmask 255.255.0.0 broadcast 10.6.255.255
    inet6 fe80::316:9102:e23b:96d6 prefixlen 64 scopeid 0x20<link>
    ether 28:c5:c8:75:d7:21 txqueuelen 1000 (Ethernet)
    RX packets 980010 bytes 1354722674 (1.3 GB)
    RX errors 0 dropped 366 overruns 0 frame 0
    TX packets 176450 bytes 19981104 (19.9 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
    device interrupt 19 memory 0x80900000-80920000

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 3134 bytes 412341 (412.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 3134 bytes 412341 (412.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

```

- **ifconfig**

ifconfig (Interface Configuration)

Purpose: Displays or configures network interface parameters.

Observation Focus: Identifies active network cards (eth0, wlan0, lo). Note the inet address (your local IPv4 address), netmask, and RX/TX packets to verify that data is actively being sent and received.

```
=====
COMMAND 3 / 16: iwconfig
RUNNING: iwconfig
=====

lo          no wireless extensions.

eno1        no wireless extensions.

=====
[ SCREENSHOT BUFFER: 5 SECONDS REMAINING ]
=====
```

- **iwconfig**

iwconfig (Wireless Interface Configuration)

- **Purpose:** Dedicated specifically to wireless network interfaces.
- **Observation Focus:** If you are on Wi-Fi, it reveals the SSID (network name), link quality, signal strength (in dBm), and data bit rate. If you are on a wired VM, it will say "no wireless extensions".

- ping

```
=====
COMMAND 4 / 16: ping
RUNNING: ping -c 4 google.com
=====

PING google.com (192.178.193.139) 56(84) bytes of data.
64 bytes from ly-in-f139.1e100.net (192.178.193.139): icmp_seq=1 ttl=113 tim
64 bytes from ly-in-f139.1e100.net (192.178.193.139): icmp_seq=2 ttl=113 tim
64 bytes from ly-in-f139.1e100.net (192.178.193.139): icmp_seq=3 ttl=113 tim
64 bytes from ly-in-f139.1e100.net (192.178.193.139): icmp_seq=4 ttl=113 tim

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 33.846/36.982/46.301/5.380 ms

=====
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=====
```

ping (Packet InterNet Groper)

- **Purpose:** Tests reachability of a host across an IP network and measures round-trip time.
- **Observation Focus:** Look at the time=XX ms value; lower values mean lower latency. Check the packet loss percentage at the bottom; 0% loss confirms a highly stable, working connection to the target server.

```

=====
COMMAND 5 / 16: route
RUNNING: route -n
=====

Kernel IP routing table
Destination      Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0          10.6.0.1        0.0.0.0         UG    100    0      0 eno1
10.6.0.0         0.0.0.0         255.255.0.0     U     100    0      0 eno1
169.254.0.0      0.0.0.0         255.255.0.0     U     1000   0      0 eno1
=====
[ SCREENSHOT BUFFER: 5 SECONDS REMAINING ]
=====

```

- **route**

route (Routing Table)

- **Purpose:** Displays and manipulates the IP routing table.
- **Observation Focus:** Look for the destination **0 . 0 . 0 . 0** with the **G** (Gateway) flag. This row displays your **Default Gateway** (typically your local router's IP), which is the exit point your traffic uses to reach the broader internet.

- **nmap**

```
=====
COMMAND 6 / 16: nmap
RUNNING: nmap -F localhost
=====

Starting Nmap 7.80 ( https://nmap.org ) at 2026-06-25 10:49 IST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000088s latency).
Not shown: 98 closed ports
PORT      STATE SERVICE
631/tcp   open ipp
3306/tcp  open mysql

Nmap done: 1 IP address (1 host up) scanned in 0.04 seconds

=====
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=====
```

nmap (Network Mapper)

- **Purpose:** Network exploration tool and security/port scanner.
- **Observation Focus:** The output displays a list of open ports on the target machine (`localhost`). It tells you which services (like SSH, HTTP, or RPC) are actively running and exposed on that machine.

```
=====
COMMAND 7 / 16: tcpdump
RUNNING: tcpdump -c 5 -i any
=====

tcpdump: any: You don't have permission to capture on that device
(socket: Operation not permitted)

=====
[ SCREENSHOT BUFFER: 5 SECONDS REMAINING ]
=====
[ ]
```

- **tcpdump**

tcpdump (Command-line Packet Analyzer)

- **Purpose:** Captures and displays network packets flowing through an interface in real-time.
- **Observation Focus:** Each line represents a packet. Observe the timestamps, source IP/port, destination IP/port, and protocol flags (like [S] for SYN or [.] for ACK), showcasing the underlying handshake mechanics of network traffic.

- dig

```

=====
COMMAND 8 / 16: dig
RUNNING: dig google.com
=====

; <<>> DiG 9.18.28-0ubuntu0.22.04.1-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 8692
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:;; udp: 65494
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                 61      IN      A      192.178.193.139
google.com.                 61      IN      A      192.178.193.102
google.com.                 61      IN      A      192.178.193.100
google.com.                 61      IN      A      192.178.193.101
google.com.                 61      IN      A      192.178.193.138
google.com.                 61      IN      A      192.178.193.113

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Thu Jun 25 10:50:08 IST 2026
;; MSG SIZE rcvd: 135

```

dig
(Domain Information Groper)

- Purpose: A flexible tool for interrogating DNS name servers.

- Observation Focus: Look at

the ANSWER SECTION. It shows the mapping of a human-readable domain name to its direct public IPv4 address, along with the TTL (Time to Live) value.

- traceroute

```

=====
COMMAND 9 / 16: traceroute
RUNNING: traceroute -m 15 google.com
=====

traceroute to google.com (192.178.193.100), 15 hops max, 60 byte packets
 0  _gateway (10.6.0.1)  0.461 ms  0.435 ms  0.470 ms
 1  10.106.0.2 (10.106.0.2)  0.405 ms  0.391 ms  0.380 ms
 2  10.25.1.10 (10.25.1.10)  0.318 ms  0.306 ms  0.295 ms
 3  * * 122.15.35.174 (122.15.35.174)  0.921 ms
 4  182.79.198.2 (182.79.198.2)  5.373 ms 182.79.236.125 (182.79.236.125)  5.506 ms  5.494 ms
 5  192.178.121.25 (192.178.121.25)  1.907 ms 192.178.121.23 (192.178.121.23)  3.253 ms  3.052 ms
 6  * 142.250.239.228 (142.250.239.228)  2.860 ms *
 7  142.251.55.40 (142.251.55.40)  5.549 ms 216.239.49.109 (216.239.49.109)  20.715 ms  20.423 ms
 8  142.250.208.152 (142.250.208.152)  17.240 ms 142.250.239.56 (142.250.239.56)  6.427 ms 142.250.239.228 (142.250.239.228)  6.747 ms
 9  142.251.241.173 (142.251.241.173)  34.140 ms 108.170.236.181 (108.170.236.181)  25.487 ms 192.178.82.107 (192.178.82.107)  25.472 ms
10  192.178.254.220 (192.178.254.220)  29.999 ms 142.251.245.192 (142.251.245.192)  24.987 ms 142.251.245.194 (142.251.245.194)  24.240 ms
11  142.250.59.103 (142.250.59.103)  29.512 ms 142.251.65.155 (142.251.65.155)  27.196 ms 142.251.78.205 (142.251.78.205)  28.357 ms
12  * 192.178.45.12 (192.178.45.12)  28.325 ms *
13  * * *
14  * * *
15  * * *

[ SCREENSHOT BUFFER: 5 SECONDS REMAINING ]
=====

```

traceroute (Trace Route)

- Purpose: Displays the route (path) and measures transit delays of packets across an IP network.
- Observation Focus: Each numbered line represents a "hop" (a router) between your computer and the target destination. If you see asterisks (* * *), it means that specific router is configured to drop diagnostic ICMP/UDP requests for security.

- mtr

mtr (My Traceroute)

- **Purpose:** Combines the functionality of ping and traceroute into a continuous diagnostic tool.
- **Observation Focus:** The report mode prints out a summary table showing every hop along the path alongside real-time stats like Loss%, Best, Avg, and Worst latency timings, making it easy to isolate exactly which router is causing a bottleneck.

- curl

```
COMMAND 11 / 16: curl
RUNNING: curl -I https://www.google.com
=====
HTTP/2 429
content-length: 1103
content-type: text/html; charset=UTF-8
date: Thu, 25 Jun 2026 05:20:46 GMT
alt-svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000
[ SCREENSHOT BUFFER: 5 SECONDS REMAINING ]
=====
6. [ 142.123.147.230 00.0% 3 7.4 7.4 7.4 7.4 0.0
7. | -- 142.250.208.105 0.0% 3 7.8 7.6 7.1 7.8 0.4
8. | -- 142.251.230.52 0.0% 3 6.6 6.6 6.4 6.8 0.2
9. | -- ??? 100.0 3 0.0 0.0 0.0 0.0 0.0
10. | -- 192.178.254.224 0.0% 3 43.0 35.2 29.7 43.0 6.9
11. | -- 142.250.211.201 0.0% 3 30.7 30.7 30.7 30.8 0.1
12. | -- 142.251.245.194 0.0% 3 30.3 30.3 30.3 30.4 0.0
13. | -- ??? 100.0 3 0.0 0.0 0.0 0.0 0.0
14. | -- ??? 100.0 3 0.0 0.0 0.0 0.0 0.0
15. | -- ??? 100.0 3 0.0 0.0 0.0 0.0 0.0
16. | -- ??? 100.0 3 0.0 0.0 0.0 0.0 0.0
17. | -- ??? 100.0 3 0.0 0.0 0.0 0.0 0.0
18. | -- ??? 100.0 3 0.0 0.0 0.0 0.0 0.0
19. | -- ly-in-f139.1e100.net 0.0% 3 33.9 34.6 33.9 35.9 1.1
[ SCREENSHOT BUFFER: 5 SECONDS REMAINING ]
=====
```

curl
(Client
URL)

- **Purpose:** Transfers data to or from a server.

- **Obs**

ervation Focus: Using the -I flag fetches only the HTTP response headers. Observe the HTTP/2 200 OK (or 301/302 Redirect) status code, the Server type, and cache configurations, proving successful application-layer communication.

- wget

```
=====
COMMAND 12 / 16: wget
RUNNING: wget --spider https://www.google.com
=====

spider mode enabled. Check if remote file exists.
-2026-06-25 10:50:51-- https://www.google.com/
Resolving www.google.com (www.google.com)... 142.251.156.119, 142.251.153.119, 142.251.151.119, ...
Connecting to www.google.com (www.google.com)|142.251.156.119|:443... connected.
HTTP request sent, awaiting response... 204 No Content
Remote file exists.

=====
[ SCREENSHOT BUFFER: 5 SECONDS REMAINING ]
=====
]
```

wget (World Wide Web Get)

- **Purpose:** Non-interactive network downloader.
- **Observation Focus:** The `--spider` flag verifies the remote file or page exists without downloading it. The output shows the connection handshake, HTTP request sent, and the remote server's response code.

• `tracpath`

```
=====
COMMAND 13 / 16: tracpath
RUNNING: tracpath -m 15 google.com
=====

1?: [LOCALHOST]                pmtu 1500
1:  _gateway                    0.797ms asymm  2
1:  _gateway                    0.642ms asymm  2
2:  10.106.0.2                  0.551ms asymm  3
3:  10.25.1.10                  0.417ms
4:  ???                        5.317ms
5:  ???                        5.943ms asymm  6
6:  no reply
7:  no reply
8:  no reply
9:  no reply
10: no reply
11: no reply
12: no reply
13: no reply
14: no reply
15: no reply
Too many hops: pmtu 1500
Resume: pmtu 1500
```

tracert (Trace Path)

- **Purpose:** Traces the network path to a remote host, similar to traceroute, but maps the MTU (Maximum Transmission Unit).
- **Observation Focus:** It determines the path MTU size (usually 1500 bytes), highlighting whether packets are forcing fragmentation down the line.

```
=====
COMMAND 14 / 16: nslookup
RUNNING: nslookup google.com
=====

Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 192.178.193.113
Name:   google.com
Address: 192.178.193.138
Name:   google.com
Address: 192.178.193.139
Name:   google.com
Address: 192.178.193.100
Name:   google.com
Address: 192.178.193.101
Name:   google.com
Address: 192.178.193.102
Name:   google.com
Address: 2404:6800:4007:830::200e
```

- nslookup

nslookup (Name Server Lookup)

- **Purpose:** Queries internet name servers interactively or non-interactively.
- **Observation Focus:** Displays the IP address of the local DNS server used to resolve the query, followed by the "Non-authoritative answer" containing the IP addresses associated with the target domain.

```
=====
COMMAND 15 / 16: host
RUNNING: host google.com
=====

google.com has address 192.178.193.113
google.com has address 192.178.193.101
google.com has address 192.178.193.138
google.com has address 192.178.193.102
google.com has address 192.178.193.139
google.com has address 192.178.193.100
google.com has IPv6 address 2404:6800:4007:830::200e
google.com mail is handled by 10 smtp.google.com.
```

- **host**

host (DNS Lookup Utility)

- **Purpose:** A simple, concise utility for performing DNS lookups.
- **Observation Focus:** Provides a highly readable, short summary showing that the domain name translates directly to its corresponding IPv4 and IPv6 addresses, as well as its mail handling servers (MX records).

- **whois**

```
=====
COMMAND 16 / 16: whois
RUNNING: whois google.com | head -n 30
=====

Domain Name: GOOGLE.COM
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2019-09-09T15:39:04Z
Creation Date: 1997-09-15T04:00:00Z
Registry Expiry Date: 2028-09-14T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1.GOOGLE.COM
Name Server: NS2.GOOGLE.COM
Name Server: NS3.GOOGLE.COM
Name Server: NS4.GOOGLE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-06-25T05:21:32Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
```

whois (Domain Registry Lookup)

- **Purpose:** Queries databases that store the registered users or assignees of an internet resource.
- **Observation Focus:** Displays public ownership records of the domain, including the domain registrar, creation date, expiration date, and assigned administrative name servers.